

Problem Solving in Programming

Quiz | 15 Multiple Choice Questions

Name: _____ Date: _____

1. A programming language such as C, C++, Java, or Python is best described as:

- A) The solution to every problem
- B) Only a tool; understanding the problem still matters most
- C) A replacement for problem analysis
- D) A method of testing

2. Which of the following best defines a "problem" in the context of problem solving?

- A) A piece of code that fails to compile
- B) A situation or task where a goal exists but the steps to reach it are unknown
- C) A syntax error in a program
- D) A finished algorithm

3. What is programming, at its core, mainly about?

- A) Memorizing syntax
- B) Converting a real problem into a sequence of precise instructions a computer can execute
- C) Writing comments
- D) Testing hardware

4. What is the correct order of the general problem-solving process shown in the material?

- A) Algorithm → Problem Definition → Analysis → Strategy
- B) Problem Definition → Analysis → Strategy → Algorithm → Flowchart/Pseudocode → Code → Testing → Solution
- C) Code → Testing → Analysis → Solution
- D) Strategy → Code → Problem Definition → Testing

5. Which question does Problem Definition primarily answer?

- A) HOW is the problem structured?
- B) WHAT exactly are we trying to solve?
- C) WHEN should the code run?
- D) WHO will use the program?

6. Which of the following is NOT listed as part of a good problem definition?

- A) The goal or objective
- B) A basic description of required input and expected output

- C) The exact source code for the solution
- D) Important requirements or boundaries

7. Which statement is an example of a CLEARER problem definition, according to the material?

- A) "Students are getting low marks."
- B) "Determine the average marks of a student from five subject marks."
- C) "Fix the grading system."
- D) "Marks need improvement."

8. Problem Analysis mainly investigates:

- A) WHAT the problem is
- B) WHO created the problem
- C) HOW the problem is structured — inputs, outputs, processing, conditions, constraints, and smaller tasks
- D) WHEN the problem occurred

9. In the "Calculate the Average of 5 marks" example, what is the correct order of steps?

- A) Divide total by 5 → Get 5 marks → Add them → Display average
- B) Get 5 marks → Add them → Divide total by 5 → Display average
- C) Display average → Add them → Get 5 marks → Divide total by 5
- D) Add them → Get 5 marks → Display average → Divide total by 5

10. Why is it useful to break a large problem into smaller parts?

- A) It makes the problem harder to test
- B) Smaller parts are easier to understand, design, test, and implement than one large task
- C) It increases ambiguity
- D) It removes the need for an algorithm

11. Which problem-solving strategy is described as "breaking a large program into smaller functions"?

- A) Trial & Error
- B) Working Backward
- C) Divide & Conquer
- D) Brute Force

12. "Solving a maze from the exit toward the entrance" is an example of which strategy?

- A) Pattern Recognition
- B) Working Backward
- C) Analogy

D) Brute Force

13. Recognizing that the sequence 2, 4, 6, 8 follows a +2 pattern is an example of which strategy?

- A) Pattern Recognition
- B) Divide & Conquer
- C) Trial & Error
- D) Working Backward

14. An algorithm is best defined as:

- A) Any block of source code
- B) A finite sequence of clear, step-by-step instructions to solve a problem
- C) A flowchart with no steps
- D) A programming language

15. According to the material, can an algorithm have 1000 steps?

- A) No, algorithms must always be short
- B) Yes, as long as the steps are finite, clear, and effective
- C) No, more than 10 steps is not allowed
- D) Only if it uses a loop